

Hyunwoo Kwon (Will Kwon)

Ph. D. student
Division of Applied Mathematics
Brown University
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Education

Brown University

Ph. D. candidate in the Division of Applied Mathematics

Providence, RI, USA

Aug. 2022 – Present

Thesis Advisor : Prof. Hongjie Dong

Brown University

Sc. M. in Applied Mathematics

Providence, RI, USA

Aug. 2022 – May. 2024

Sogang University

M.S. in Mathematics

Seoul, Korea

Feb. 2017 – Feb. 2019

Thesis Advisor : Prof. Hyunseok Kim

Thesis: Dirichlet and Neumann problems for elliptic equations with singular drifts on Lipschitz domains

Sogang University

B.S. in Mathematics, *Summa Cum Laude*

Seoul, Korea

Feb. 2013 – Feb. 2017

Employment

Researcher

Sogang University

Seoul, Korea

Jun. 2022 – Jul. 2022

Assistant Professor with the rank First Lieutenant

Republic of Korea Air Force Academy

Cheongju, Korea

Sep. 2020 – May. 2022

Lecturer with the rank Second Lieutenant

Republic of Korea Air Force Academy

Cheongju, Korea

Aug. 2019 – Sep. 2020

Research Interests

Partial differential equations, Real variable methods in harmonic analysis

Publication and Preprints

1. *Global well-posedness of the one-phase Muskat problem with surface tension*, with H. Dong, [arXiv:2604.06545 \[math.AP\]](#)
2. *Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential*, with W. Huang, [arXiv:2602.21344 \[math.AP\]](#)
3. *Recent progress on local regularity estimates for the nonstationary Stokes equations*, with H. Dong, to appear in AMS Contemporary Mathematics.
4. *Spatial C^1 , C^2 , and Schauder estimates for Nonstationary Stokes equations with Dini mean oscillation coefficients*, with H. Dong, *Calc. Var.* 65, (2026), 43, [arXiv:2412.20125 \[math.AP\]](#)
5. *Nonstationary Stokes equations on a domain with curved boundary under slip boundary conditions*, with H. Dong, *Discrete Contin. Dyn. Syst.*, [arXiv:2408.17321 \[math.AP\]](#)
6. *Scattering of the Vlasov-Riesz system in the three dimensions*, with W. Huang, [arXiv:2407.16919 \[math.AP\]](#)
7. *Local and global solutions to Stokes-Magneto equations with fractional dissipations*, with H. Bae and J. Shin, *Partial Differ. Equ. Appl.* 6, (2025), 32, [arXiv:2310.03255 \[math.AP\]](#)
8. *Interior and boundary mixed derivative estimates for nonstationary Stokes equations*, with H. Dong, *Commun. Contemp. Math.* (2025) 2550069, [arXiv:2308.09220 \[math.AP\]](#)
9. *Global existence and uniqueness of weak solutions of a Stokes-Magneto system with fractional diffusions*, with H. Kim, *J. Differential Equations* 374(2023), 497–547, [arXiv:2302.02046 \[math.AP\]](#)
10. *Elliptic equations in divergence form with drifts in L^2* , *Proc. Amer. Math. Soc.* 150(2022), no. 8, 3415–3429, [arXiv:2104.01300 \[math.AP\]](#)
11. *Existence and uniqueness of weak solution in $W^{1,2+\varepsilon}$ for elliptic equation with drifts in weak- L^n spaces*, *J. Math. Anal. Appl.* 500(1) (2021) 125165, [arXiv:2011.07524 \[math.AP\]](#)
12. *Dirichlet and Neumann problems for elliptic equations with singular drifts on Lipschitz domains*, with H. Kim, *Trans. Amer. Math. Soc.* 375 (2022), 6537–6574, [arXiv:1811.12619 \[math.AP\]](#)

Awards, Grants & Honours

ICM 2026 travel grant (part of NSF award #2620718, \$2,400)	2026
BK Plus Scholarship, Sogang University	2017–2019
Albatross Fellowship (full funding support, 2 years), Sogang University	2017–2019
President prize for highest achievement in academic(1%), Sogang University	2015
Dean's List of Natural Science, Sogang University for highest achievement in academic	2014
Full Funding Scholarship from Hyundai Chung Mong-Koo Foundation	2013–2017

Teaching

Instructor

Brown University

Providence, RI, USA

Jun. 2025 – Aug. 2025

- 2025-Summer APMA 1210 Operations Research: deterministic models

Recitation Instructor and Grader

Brown University

Providence, RI, USA

Sep. 2023 – Present

- 2026-Spring APMA 0360 Applied Partial Differential Equation (Dr. Ryan Creedon)
- 2025-Fall APMA 0360 Applied Partial Differential Equation (Prof. Yan Guo)
- 2025-Summer APMA Bridge Course in Analysis (Dr. Peyam Tabrizian)
- 2025-Spring APMA 0350 Applied Ordinary Differential Equation (Dr. Zhuolun Yang)
- 2024-Summer APMA Bridge Course in Analysis (Dr. Peyam Tabrizian)
- 2024-Spring APMA 0360 Applied Partial Differential Equations I (Dr. Timur Yastrzhembskiy)
- 2023-Fall APMA 1210 Operations Research: deterministic models (Dr. Amalia Culiuc)

Lecturer

Republic of Korea Air Force Academy

Cheongju, Korea

Sep. 2019 – May. 2022.

- 2022-1 Calculus (2 classes), Engineering Mathematics II
- 2021-2 Linear Algebra (Humanities, 2 classes), (Science, 1 class)
- 2021-1 Calculus (Science, 2 classes), Engineering Mathematics I
- 2020-2 Linear Algebra (Humanities), Engineering Mathematics I, Engineering Mathematics II
- 2020-1 Calculus (Humanities), Engineering Mathematics I (2 classes)
- 2019-2 Linear Algebra (Science, 3 classes), Engineering Mathematics I

Coordinator of Undergraduate Seminar in Sogang Math Honors Club

Sogang University

Seoul, Korea

Apr. 2017 – Aug. 2017

- 2017-1 Introduction to Hilbert Spaces

Recitation Instructor and Grader

Sogang University

Seoul, Korea

Mar. 2017 – Feb. 2019

- 2018-2 Advanced calculus II (Recitation instructor, Grading)
- 2018-1 Advanced calculus I, Theory of complex variable I (Grading)
- 2017-2 Advanced calculus II (Recitation instructor, Grading)
- 2017-1 Advanced calculus I (Recitation instructor)

Mathematics Tutor

Sogang University

Seoul, Korea

Mar. 2016 – Dec. 2016

- Worked as a tutor at Sogang Mathematics Learning Center
- Helped several subjects which use mathematics

Mentoring Undergraduate Students

Lucas Sha, Variational approach to kinetic problems, 2026 Summer independent study, Brown University

Lloyd Sangwoo Ko and Lucas Sha, Fourier Analysis, 2026 Spring independent study, Brown University

Charlotte Kouroriez and Joseph Han, Well-posedness of KdV equation, 2025 Fall DRP, Brown University

Yongxi Lin, Introduction to the mathematical theory on incompressible fluids, 2025 Summer independent study, Brown University

Lloyd Sangwoo Ko, Lean Quimbo, Long time behavior of fluids, 2025 Spring DRP, Brown University

Talks in Mathematics

Contribution talks, ICM 2026	Jul. 2026
Current Trends in Fluid Dynamics, University of Maryland	Apr. 2026
Special Session on Fluid dynamics and nonlinear partial differential equations, 2026 AMS Spring Southeastern Sectional Meeting	Mar. 2026
KIAS Analysis, PDE&Probability Seminar (online)	Feb. 2026
Poster session in Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburgh	Nov. 2025
PDE seminar, Brown University	Sep. 2025
Poster session in Theory and Applications of Elliptic PDE, UC Irvine	Jun. 2025
Poster session in Workshop for Research and Workforce Development in Fluid Mechanics, University of Nebraska-Lincoln	May 2025
Special Session on Recent developments in nonlinear evolution equations, Spring 2025 AMS Eastern Sectional Meeting	Apr. 2025
Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook University	Jan. 2025
PDE and Applied Analysis Seminar, POSTECH (online)	Sep. 2024
Poster session in Spectral Analysis of Schrödinger operators, ICERM	Aug. 2024
PDE seminar, Yonsei University (online)	Aug. 2024
Nonlinear PDEs Summer School, UC Berkeley	Jun. 2024
Poster session in Rivière-Fabes Symposium on Analysis and PDE, University of Minnesota	Apr. 2024
2024 AMS New England Graduate Student Conference	Apr. 2024
BK Seminar for Graduate Students, Yonsei University	Mar. 2024
AMS Special Session on Dynamics and Regularity of PDEs, 2024 Joint Mathematical Meeting	Jan. 2024
Analysis seminar, Temple University	Nov. 2023
Poster session in Prairie Analysis Seminar, Kansas State University	Nov. 2023
PDE and Differential Geometry seminar, University of Connecticut	Oct. 2023
PDE & Applied Analysis Seminar, Sogang University	Jun. 2023
Special Session on Dynamics of Partial Differential Equations, 2023 AMS Spring Southeastern Sectional Meeting	Mar. 2023

PDE seminar, UNIST	Jun. 2022
PDE and Applied Analysis Seminar, POSTECH	Jun. 2022
Contribution talks in Analysis session, 2022 Spring Meeting of KMS (online)	Apr. 2022
Workshop for young mathematicians in Korea (online)	Jan. 2022
Contribution talks in Analysis session, 2021 Spring Meeting of KMS (online)	Apr. 2021
KIAS Analysis, PDE & Probability Seminar (online)	Jan. 2021
HA-GMT-PDE seminar at University of Minnesota (online)	Sep. 2020
KIAS Analysis, PDE & Probability Seminar	Dec. 2019
Contribution talks in Analysis session, 2019 Spring Meeting of KMS	Oct. 2019
IBS-CGP Seminar	Jan. 2019
KIAS Analysis&PDE Learning seminar	Oct. 2018
Contribution talks in Analysis session, 2018 Joint Meeting of KMS and DMV	Oct. 2018
Graduate student seminar, Yonsei University	Sep. 2018
Workshop for students in Analysis and PDE	Jul. 2018

Work Experience

L^AT_EX Technician

Freelancer

Seoul, Korea

Mar. 2014 – Feb. 2019

- Made several books with Kyungmoon Publisher and the others
- Technical Editor for Publication of ICM 2014 Seoul, Korea

Academic Services

APMA committee

- Graduate Student Representative in APMA (2024.8–2025.8)
- Graduate Student Council member as a representative of APMA (2024–)
- DEI committee member (graduate student subcommittee) (2024.8–2025.8)

Journal Referee

DCDS-B, J. Math. Anal. Appl., Comm. Pure. Appl. Anal., J. Differential Equations, Comm. PDEs(2), Math. Ann., Acta Math. Scientica, Adv. Math.

Organizer of learning seminar

Recent developments in surface water waves

Providence, RI, USA

Sep. 2024 – Dec. 2024

Organizer of seminar

Graduate Lecture series in Analysis and PDEs (GLESPEA)

Providence, RI, USA

Jan. 2023 – May 2026

Co-organized with T. G. Borges, F. Benito Fernández de la Cigoña, M. Pasquariello: [homepage](#)

Organizer of learning seminar

Harmonic analysis related to PDEs

Providence, RI, USA

Sep. 2023 – Dec. 2023

Organizer of workshop

Workshop for young mathematicians in Korea

Cheongju, Korea

Jan. 2022

Co-organized with D.-Y. Ko (Rutgers Univ.) and B. Sohn (SNU) : [homepage](#)

Organizer of workshop

Workshop for students in Analysis and PDE

Seoul, Korea

Jul. 2018

Co-organized with H. Kim (Sogang Univ.), Y.-R. Lee (Sogang Univ.) and S. W. Ryoo (SNU) : [homepage](#)

Founder and Organizer of Undergraduate Mathematics Seminar

United Mathematics club in Korea : [Stupid Mathematics Seminar](#)

Seoul, Korea

Feb. 2014 – Feb. 2016

Conference and Workshop Attended (will attend)

- Partial Differential Equations for Many Particles Systems, ICERM, RI, USA, Aug. 6 – Aug. 10, 2026.
- International Congress of Mathematicians 2026, PA, USA, Jul. 23 – Jul. 30, 2026.
- Nonlinear waves: Nonlinear Waves: From Theoretical and Computational Advances to Experimental Observations, ICERM, RI, USA, Apr. 27 – May. 1, 2026.
- Current Trends in Fluid Dynamics, University of Maryland, MD, USA, Apr. 8 – Apr. 10, 2026.
- 2026 AMS Spring Southeastern Sectional Meeting, Georgia Tech, Georgia Southern University-Armstrong Campus, GA, USA, Mar. 28 – Mar. 29, 2023.
- Workshop in PDE and Applied Math in the Northeast Region, University of Connecticut, CT, USA, Mar. 7, 2026.
- Fluid Dynamics and Turbulence, ICERM, RI, USA, Nov. 22 – Nov. 23, 2025.
- Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburgh, PA, USA, Nov. 14 – Nov. 16, 2025.
- Theory and Applications of Elliptic PDE, UC Irvine, CA, USA, Jun. 23 – Jun. 27, 2025.
- Workshop for Research and Workforce Development in Fluid Mechanics, University of Nebraska-Lincoln, NE, USA, May 9 – May 11, 2025.
- 2025 AMS Spring Eastern Sectional Meeting, University of Connecticut, CT, USA, Apr. 5 – Apr. 6, 2025.
- Workshop on Kinetic Theory and Fluids, University of Wisconsin-Madison, WI, USA, Mar. 28 – Mar. 30, 2025.
- Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook University, NY, USA, Jan. 6 – Jan. 10, 2025.
- Spectral Analysis of Schrödinger Operators, ICERM, RI, USA, Aug. 19 – Aug. 23, 2024.
- Nonlinear PDEs Summer School at UC Berkeley, UC Berkeley, CA, USA, Jun. 12 – Jun. 14, 2024.
- Recent Advances in Nonlinear Partial Differential Equations, University of Minnesota, MN, USA, May 13 – May 17, 2024.
- Rivière-Fabes Symposium on Analysis and PDE, University of Minnesota, MN, USA, Apr. 19 – Apr. 21, 2024.
- AMS New England AMS Grad Student Conference, Brown University, RI, USA, Apr. 13, 2024.

- Fluids in Seoul 2024, KIAS, Seoul, Korea, Jan. 15 – Jan. 19, 2024.
- Joint Mathematics Meetings 2024, San Fransisco, CA, USA, Jan. 3 – Jan 6, 2024.
- 19th Prairie Analysis Seminar, Kansas State University, KS, USA, Nov. 3 – Nov. 4, 2023.
- Summer School: Fluid Dynamics, University of Maryland, MD, USA, Jul. 17 – Jul. 21, 2023.
- PDE in Seoul 2023: Conference in honor of Professor Mikhail Safonov's 70th birthday, Yonsei University, Seoul, Korea, Jun. 19 – Jun. 23, 2023.
- 2023 AMS Spring Southeastern Sectional Meeting, Georgia Tech, Atlanta, GA, USA, Mar. 18 – Mar. 19, 2023.
- Joint Mathematics Meetings 2023, Boston, MA, USA, Jan. 4 – Jan 7, 2023.
- Intensive lectures on the Cauchy problems for quasilinear PDEs by Prof. Sung-Jin Oh, Online via Zoom, Jul. 26, 2021 – Jul. 30, 2021.
- Yonsei Ewha Sogang Seminar on Analysis & PDEs, Online via Zoom, Jan. 20, 2021.
- Workshop on Harmonic Analysis Techniques for PDEs, JBNU, Jeonju, Korea, Feb. 13, 2019 – Feb. 14, 2019.
- The 3rd meeting of Young PDE researchers in PDEs, KIAS, Seoul, Korea, Dec. 13, 2018 – Dec. 14, 2018.
- Workshop for students in Analysis and PDE(as a orgainzer and speaker), Sogang Univ., Seoul, Korea, Jul. 12, 2018 – Jul. 13, 2018.
- Dynamics of partial differential equations, KAIST, Daejeon, Jun. 27, 2018 – Jun. 29, 2018.
- 2018 Korea PDE School #8, NIMS, Daejeon, Jan. 29, 2018 – Feb. 02, 2018.
- Minicourse: General relativity in spherical symmetry, KIAS, Apr. 07, 2017 – Apr. 28, 2017.
- Lecture Series on various topics in PDEs, KIAS, Seoul, Korea, Feb. 13, 2017 – Feb. 17, 2017 / Feb. 20, 2017 – Feb. 24, 2017.
- 2017 Korea PDE School #7, NIMS, Daejeon, Jan. 16, 2017 – Jan. 20, 2017.
- 2016 CMC Foundations for Mathematical Challenge: Solitons in the Korteweg-de Vries equations, inverse scattering approaches, KIAS, Seoul, Korea, Aug. 22, 2016 – Aug. 26, 2016.
- IBS-CGP Mathematics Festival 2016, IBS Center for Geometry and Physics, Pohang, Jul. 18, 2016 – Jul. 22, 2016.
- 7th Workshop on PDEs of Mathematical Physics, UNIST, Ulsan, Jan. 14, 2016 – Jan. 16, 2016.

Updated: June 3, 2026